Concavity and Initialization for Unsupervised Dependency Grammar Induction

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Abstract

We examine models for unsupervised learning with concave log-likelihood functions. We begin with the most well-known example, IBM Model 1 for word alignment (Brown et al., 1993), and study its properties, discussing why other models for unsupervised learning are so seldom concave. We then present concave models for dependency grammar induction and validate them experimentally. Despite their simplicity, we find that initializing the dependency model with valence using our concave models can approach state of the art grammar induction results for English and Chinese.

1 Introduction

In NLP, unsupervised learning typically implies optimization of an extremely bumpy objective function riddled with local maxima. However, one exception is IBM Model 1 (Brown et al., 1993) for word alignment, which is the only model commonly used for unsupervised learning in NLP that has a concave log-likelihood function (though not strictly concave (Toutanova and Galley, 2011)). For other applications, such as unsupervised part-of-speech tagging and grammar induction, and indeed for more sophisticated word alignment models, the log-likelihood function optimized by EM is non-concave. As a result, researchers are obligated to consider initialization in addition to model design (Klein and Manning, 2004; Goldberg et al., 2008).

For example, consider the dependency grammar induction results shown in Table 1 when training the widely used dependency model with valence (DMV; Klein and Manning, 2004). Using uniform distributions for initialization (UNIF) results in an accuracy of 17.6% on the test set, well below the baseline of attaching each word to its right neighbor (ATTACHRIGHT, 31.7%). Furthermore, when using a set of 50 random initializers (RAND), the standard deviation of the accuracy is an alarming 8.3%.

In light of this sensitivity to initialization, it is compelling to consider unsupervised models with concave log-likelihood functions. In this paper, we explore the issues involved with such an expedition and elucidate the limitations of such models for unsupervised NLP. We describe simple concave models for dependency grammar induction that are easy to implement and offer efficient optimization. We use our models to initialize the DMV and find that they achieve surprisingly competitive performance for English and Chinese, even rivalling that of much more sophisticated methods.

2 IBM Model 1 and Concavity

IBM Model 1 is a conditional model of a target-language sentence $\boldsymbol{e}$ of length m and an alignment $\boldsymbol{a}$ given a source-language sentence $\boldsymbol{f}$ of length l. The generation of m is assumed to occur with some (inconsequential) uniform probability ϵ. The alignment vector $\boldsymbol{a}$ has an entry for each element of $\boldsymbol{e}$ that contains the index in $\boldsymbol{f}$ of the aligned word. These entries are used to define which translation parameters $t(e_j|f_{a_j})$ are active. Model 1 assumes that the probability of the ith element in $\boldsymbol{a}$, denoted $a(i|j,l,m)$, is simply a uniform distribution over all l source words plus the null word. These assumptions result in the following log-likelihood for a sentence pair $\langle \boldsymbol{f}, \boldsymbol{e} \rangle$ under Model 1 (marginalizing out $\boldsymbol{a}$) (Brown et al., 1993): $\log p(\boldsymbol{e}|\boldsymbol{f})$

$$= \log \frac{\epsilon}{(l+1)^m} + \sum_{j=1}^{m} \log \sum_{i=0}^{l} t(e_j|f_i) \quad (1)$$

The only parameters to be learned in the model are $t(e_j|f_i)$. Since a parameter is concave in itself, the sum of concave functions is concave, and the log of a concave function is concave, Eq. 1 is concave in

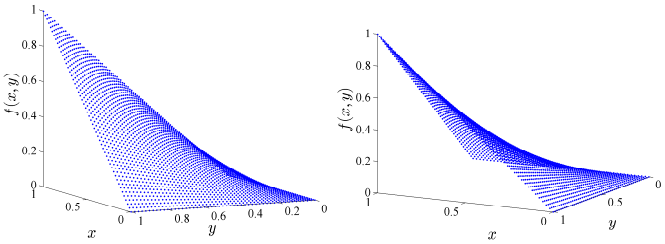

Figure 1: Plot of $f(x, y) = xy$, two views for clarity.

its parameters (Brown et al., 1993; Toutanova and Galley, 2011).

IBM Model 2 involves a slight change to Model 1 in which the probability of a word link depends on the word positions. However, this change renders it no longer concave. Consider the log-likelihood function for Model 2:

$$\log \epsilon + \sum_{j=1}^{m} \log \sum_{i=0}^{l} t(e_j|f_i)a(i|j, l, m) \quad (2)$$

Eq. 2 is not concave in the parameters $t(e_j|f_i)$ and $a(i|j, l, m)$ because a product is neither convex nor concave in its vector of operands. This can be shown by computing the Hessian of $f(x, y) = xy$ and showing that it is indefinite. A plot of $f(x, y)$ is shown in Fig. 1.

To summarize, concavity is lost when the log-likelihood function contains a product of model parameters enclosed within a $\log \sum$. If the sum is not present, the $\log$ can be used to separate the product of parameters, making the function concave. It can also be shown that a "featurized" version (Berg-Kirkpatrick et al., 2010) of Model 1 is not concave. More generally, any non-concave function enclosed within $\log \sum$ will cause the log-likelihood function to be non-concave, though there are few other non-concave functions with a probabilistic semantics than those just discussed.[1]

3 Concave, Unsupervised Models

Nearly every other model used for unsupervised learning in NLP has a non-concave log-likelihood function. Therefore, some researchers have explored convex relaxations for unsupervised learning (Guo and Schuurmans, 2007). While mathematically sophisticated, such approaches have not yielded practical benefits. We take a different approach and now proceed to describe the conditions necessary to develop concave models for two tasks.

3.1 Part-of-Speech Tagging

We will consider a standard bigram hidden Markov model for this task. Letting $\boldsymbol{y}$ denote the tag sequence for a sentence $\boldsymbol{e}$ with m tokens, the single-example log-likelihood is:

$$\log \sum_{\boldsymbol{y}} p(\text{stop}|y_m) \prod_{j=1}^{m} p(y_j|y_{j-1})p(e_j|y_j) \quad (3)$$

where y_0 is a designated "start" symbol. Unlike IBM Models 1 and 2, we cannot reverse the order of the summation and product here because the transition parameters $p(y_j|y_{j-1})$ tie the tags of different words together, causing one tag decision to affect the others. Therefore, Eq. 3 is clearly non-concave due to the presence of a product within a $\log \sum$.

However, if the tag transition probabilities $p(y_j|y_{j-1})$ are all constants and also do not depend on the previous tag y_{j-1}, then we can rewrite Eq. 3 as the following concave log-likelihood function (using $C(y)$ to denote a constant function of tag y, e.g., a fixed tag prior distribution):

$$\log C(\text{stop}) + \log \prod_{j=1}^{m} \sum_{y_j} C(y_j)p(e_j|y_j)$$

Lacking any transition modeling power, this model appears weak as a tagging model. However, we note that we can add additional conditioning information to the $p(e_j|y_j)$ distributions and retain concavity, such as previous words and tag dictionary information. Such a model might learn certain patterns about local contexts and provide an initializer for unsupervised part-of-speech tagging. Similarly, convex models for clustering (Lashkari and Golland, 2008) might serve as useful initializers as well.

3.2 Dependency Grammar Induction

To develop grammar induction models, we begin with a version of Model 1 in which a sentence $\boldsymbol{e}$ is generated from a copy of itself (denoted $\boldsymbol{e}'$): $\log p(\boldsymbol{e}|\boldsymbol{e}')$

$$= \log \frac{\epsilon}{(m+1)^m} + \sum_{j=1}^{m} \log \sum_{i=0, i \neq j}^{m} c(e_j|e'_i) \quad (4)$$

[1] Interestingly, the geometric mean function *is* concave. That is, $f(\boldsymbol{x}) = (\prod_{i=1}^{|\boldsymbol{x}|} x_i)^{\frac{1}{|\boldsymbol{x}|}}$ is concave for $\boldsymbol{x} \in R^n_{++}$ (Boyd and Vandenberghe, 2004). This observation is reminiscent of the well-known ability of annealing to smooth a non-concave objective surface by exponentiating it to a very small value.

If a word e_j is "aligned" to e'_0, e_j is a root. This is a simple child-generation model with no tree constraint. In order to preserve concavity, we are forbidden from conditioning on other parent-child assignments in the structure or including any sort of larger constraints.

However, we can condition the child distributions on additional information about $\boldsymbol{e}'$ since it is fully observed. This conditioning information may include the direction of the edge, its distance, and any properties about the words in the sentence. We found that conditioning on direction consistently improved performance: we rewrite the c distributions as $c(e_j|e'_i, I[j-i>0])$ and denote this model CCV1.

We note that we can also include constraints in the sum over possible parents and still preserve concavity. Naseem et al. (2010) found that adding parent-child constraints to a grammar induction system can improve performance dramatically. We employ one simple rule: roots are likely to be verbs. We adapt CCV1 to restrict the summation over parents to exclude e'_0 if the child word is not a verb.[2] We only employ this restriction during EM learning for sentences containing at least one verb. For sentences without verbs, we allow all words to be the root. We denote this model CCV2.

Similar to our models, Brody (2010) recently developed grammar induction models based on the IBM word alignment models. However, while our goal is to develop concave models, Brody employed Bayesian nonparametrics in his version of Model 1 and used sampling for inference and learning, causing his version of the model to no longer be concave.

4 Experiments

We ran experiments to determine how well our concave grammar induction models CCV1 and CCV2 can perform on their own and when used as initializers for the DMV. The DMV is the foundation for many state-of-the-art unsupervised grammar induction models (many of which are listed in Tab. 1), and the majority of researchers use the Klein and Manning initializer (K&M; 2004). K&M is a deterministic harmonic initializer that computes root probabilities uniformly for each word in a sentence and sets parent-child probabilities inversely proportional to their distances, then normalizes to obtain probability distributions.

Procedure We run EM for our concave models for 100 iterations (which takes less than 1 minute on sentences ≤ 10 words), then use Viterbi decoding on the test data to ensure that we find complete dependency trees. We train each DMV model for 200 iterations and use minimum Bayes risk decoding with the final model on the test data. We use several initializers for training the DMV, including K&M and the output of our concave models. The parent-child probabilities can be used directly, since CCV1 and CCV2 both condition attachments on direction, as is also done by the DMV. However, the DMV also contains stop/continue distributions for child generation. We do not have these in our concave models, so we simply initialize them uniformly.

Data We consider English and Chinese grammar induction. For English, we train on §2-21 and test on §23 in the Penn Treebank (Marcus et al., 1993). One set of experiments uses training sentences ≤ 10 words and the other uses sentences $\leq$ 20 words, punctuation removed. For Chinese, we train on §1-270 (sentences $\leq$ 10 words) and test on §271-300 of the Chinese Treebank (Xue et al., 2004). In all cases, we report accuracy on all sentences from the test set and separately for sentences ≤ 10 words.

Results Results for English are shown in Tab. 1 and Chinese results are shown in Tab. 2. We include results from work that reported results on the same test sets. For English, the constraint on sentence roots helps a great deal, as CCV2 by itself is competitive with the DMV when testing on short sentences. The true benefit of the concave models, however, appears when using them as initializers. The DMV model initialized with CCV2 achieves a substantial improvement over all other initializers. When training on sentences of length $\leq$ 20 words (bold), the performance even exceeds that of several more sophisticated models shown in the table. In particular, the 53.1% accuracy on the test set is exceeded only by that of the substantially more complex LexTSG-DMV model of Blunsom and Cohn (2010).

Most models for this task are based on the DMV in some way and most use the K&M initializer, so it

[2] For English, V* and MD tags are considered verbs. For Chinese, only V* tags are considered verbs.

		Train ≤ 10		Train ≤ 20	
		Test		Test	
Model	Init.	≤ 10	$\leq \infty$	≤ 10	$\leq \infty$
ATTACHRIGHT	N/A	38.4	31.7	38.4	31.7
CCV1	UNIF	31.4	25.6	31.0	23.7
CCV2	UNIF	43.1	28.6	43.9	27.1
DMV	UNIF	21.3	17.6	21.3	16.4
	RAND†	41.0	31.8	-	-
	K&M	44.1	32.9	51.9	37.8
	CCV1	45.3	30.9	53.9	36.7
	CCV2	54.3	43.0	**64.3**	**53.1**

		Test ≤ 10	Test $\leq \infty$
SLN	K&M	61.3	41.4
L-EVG	RAND	68.8	-
Feature DMV	K&M	63.0	-
Viterbi EM	~K&M	65.3	47.9
LexTSG-DMV	K&M	67.7	55.7
Posterior Reg.	K&M	62.0	49.1

Table 1: Directed attachment accuracies on Section 23, for short sentences (≤ 10 words) and all ($\leq \infty$). We include selected results from the literature that also evaluate on the same test set. † For RAND, accuracies are averages over 50 random initializers; σ was 10.9 for test sentences ≤ 10 and 8.3 for all. SLN = Cohen and Smith (2009), L-EVG = Headden III et al. (2009), Feature DMV = Berg-Kirkpatrick et al. (2010), Viterbi EM = Spitkovsky et al. (2010), LexTSG-DMV = Blunsom and Cohn (2010), Posterior Reg. = Gillenwater et al. (2010).

		Test Length	
Model	Init.	≤ 10	$\leq \infty$
ATTACHRIGHT	N/A	34.9	34.6
CCV1	UNIF	38.4	21.9
CCV2	UNIF	36.2	22.4
DMV	UNIF	43.9	39.8
	K&M	38.1	32.5
	CCV1	46.4	44.6
	CCV2	46.4	44.4
SLN	K&M	51.9	35.8
PGI	K&M	56.6	-

Table 2: Dependency attachment accuracies for Chinese Treebank test set, for sentences ≤ 10 words and all sentences. For training, sentences ≤ 10 words from the CTB were used. SLN = Cohen and Smith (2009), PGI = Berg-Kirkpatrick and Klein (2010).

is likely that the availability of an improved initializer for these models could lead to substantial improvement in unsupervised grammar induction.[3]

Unlike English, the use of parent constraints did not help for Chinese. However, even without parent constraints, the concave models again show impressive performance as an initializer for the DMV, performing better than the UNIF and K&M initializers. Though results lag behind state-of-the-art models for short sentences, our 44.6% accuracy on sentences of all lengths exceeds that of SLN and is, to our knowledge, the highest reported accuracy on the full CTB test set.

[3]Of the models that do not use K&M for initialization, Headden III et al. (2009) start training with many random models and choose one partly through based on an unsupervised criterion. Spitkovsky et al. (2010) use a procedure initialized with a variant of K&M followed by several stages of learning.

5 Conclusion

We have investigated concave models for unsupervised learning, elucidated the properties necessary to achieve concavity, and developed concave models for unsupervised dependency grammar induction.

While we do not yet offer a generic procedure for extracting a concave approximation from any model for unsupervised learning, our results contribute another clear data point in favor of the general methodology of staged training in unsupervised learning, and provide a simple and powerful initialization method for dependency grammar induction.

References

T. Berg-Kirkpatrick and D. Klein. 2010. Phylogenetic grammar induction. In *Proc. of ACL*.

T. Berg-Kirkpatrick, A. Bouchard-Côté, J. DeNero, and D. Klein. 2010. Painless unsupervised learning with features. In *Proc. of NAACL*.

P. Blunsom and T. Cohn. 2010. Unsupervised induction of tree substitution grammars for dependency parsing. In *Proc. of EMNLP*.

S. Boyd and L. Vandenberghe. 2004. *Convex Optimization*. Cambridge University Press.

S. Brody. 2010. It depends on the translation: Unsupervised dependency parsing via word alignment. In *Proc. of EMNLP*.

P. F. Brown, S. A. Della Pietra, V. J. Della Pietra, and R. L. Mercer. 1993. The mathematics of statistical machine translation: Parameter estimation. *Computational Linguistics*, 19(2):263–311.

S. Cohen and N. A. Smith. 2009. Shared logistic normal distributions for soft parameter tying in unsupervised grammar induction. In *Proc. of NAACL*.

J. Gillenwater, K. Ganchev, J. Graça, F. Pereira, , and B. Taskar. 2010. Posterior sparsity in dependency grammar induction. *Journal of Machine Learning Research*.

Y. Goldberg, M. Adler, and M. Elhadad. 2008. EM can find pretty good HMM POS-taggers (when given a good start). In *Proc. of ACL*.

Y. Guo and D. Schuurmans. 2007. Convex relaxations of latent variable training. In *Proc. of NIPS*.

W. P. Headden III, M. Johnson, and D. McClosky. 2009. Improving unsupervised dependency parsing with richer contexts and smoothing. In *Proc. of NAACL*.

D. Klein and C. D. Manning. 2004. Corpus-based induction of syntactic structure: Models of dependency and constituency. In *Proc. of ACL*.

D. Lashkari and P. Golland. 2008. Convex clustering with exemplar-based models. In *Proc. of NIPS*.

M. P. Marcus, B. Santorini, and M. A. Marcinkiewicz. 1993. Building a large annotated corpus of English: The Penn Treebank. *Computational Linguistics*, 19:313–330.

T. Naseem, H. Chen, R. Barzilay, and M. Johnson. 2010. Using universal linguistic knowledge to guide grammar induction. In *Proc. of EMNLP*.

V. I. Spitkovsky, H. Alshawi, D. Jurafsky, and C. D. Manning. 2010. Viterbi training improves unsupervised dependency parsing. In *Proc. of CoNLL*.

K. Toutanova and M. Galley. 2011. Why initialization matters for ibm model 1: Multiple optima and non-strict convexity. In *Proc. of ACL*.

N. Xue, F. Xia, F.-D. Chiou, and M. Palmer. 2004. The Penn Chinese Treebank: Phrase structure annotation of a large corpus. *Natural Language Engineering*, 10(4):1–30.